

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the global microbiome sequencing services market based on service, type, technology, end user, and region
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, and opportunities)
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall microbiome sequencing services market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders operating in the microbiome sequencing services market
- To forecast the revenue of the market with respect to five regional segments—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa
- To profile the key market players and comprehensively analyze their service portfolios, market positions, and core competencies²
- To track and analyze competitive developments such as services launches, expansions, agreements, collaborations, partnerships, and acquisitions in the microbiome sequencing services market

1.2 MARKET DEFINITION

Microbiome sequencing is applied to decode the genetic material of all the microorganisms in a given sample which may contain bacteria, viruses, or fungi. Whole-genome sequencing (WGS) and amplicon sequencing are the two key microbiome sequencing services. Sequencing-based detection of microorganisms exhibits different advantages over traditional approaches, some of which are mentioned below:

- Detection of anaerobic bacteria
- More cost-effective microbiome characterization of multiplex microbial samples
- Identification of non-cultivable bacteria
- Estimate the relative abundance of microbes in a sample
- Profile hundreds of microorganisms in a single analysis
- Examine complex microbiomes and microbiota communities

The microbiome sequencing services market comprises services employed for sample preparation, sequencing, library preparation, and data analysis suitable for different samples, including human microbiome samples and environmental samples.

1. Micromarkets are the further segments and subsegments of the microbiome sequencing services market.

2. Core competencies of companies are captured in terms of the key developments, market shares, and key strategies adopted to sustain their positions in the market.

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Service	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing services, including sample preparation, sequencing and library preparation, and data analysis.	NA
Type	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service types, such as amplicon sequencing, whole-genome sequencing, shotgun sequencing, and de novo sequencing.	NA
Technology	Qualitative and quantitative analysis (in terms of USD million) of technologies, such as sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and semiconductor sequencing.	NA
End User	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service end users such as pharmaceutical and biotechnology companies, academic and research institutes, small-scale CROs, and CDMOs.	NA

1.3 MARKET SCOPE

1.3.1 MARKETS COVERED

MICROBIOME SEQUENCING SERVICES MARKET			
	SERVICE - Sample Preparation - Sequencing and Library Preparation - Data Analysis		TYPE - Amplicon Sequencing - Whole-genome Sequencing - Shotgun Sequencing - Other Types
	END USER - Pharmaceutical and Biotechnology Companies - Academic and Research Institutes - Other End Users		REGION - North America (US and Canada) - Europe (Germany, UK, France, Italy, Spain, and Rest of Europe) - Asia Pacific (Japan, China, India, and Rest of Asia Pacific) - Latin America (Brazil and Rest of Latin America) - Middle East and Africa
			
TECHNOLOGY - Sequencing by Synthesis - Sequencing by Ligation - Nanopore Sequencing - Other Technologies			

Note 1: Other types of microbiome sequencing services include de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing.

Note 2: Other microbiome sequencing technologies include semiconductor sequencing and Sanger sequencing.

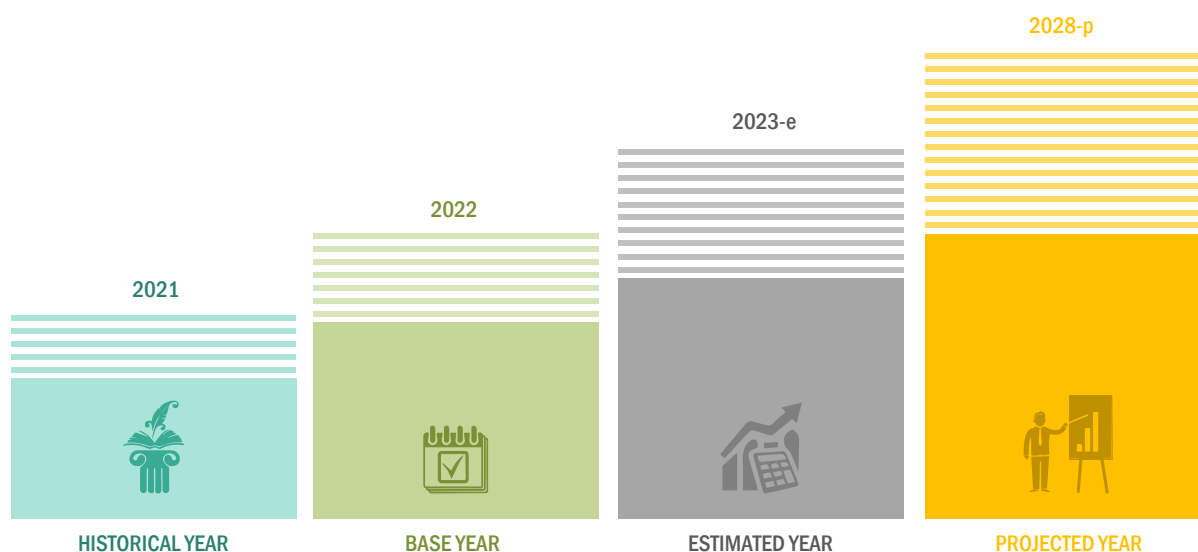
Note 3: Other end users include hospitals and clinics, small-scale CROs, and CDMOs.

Note 4: The Rest of Europe includes Switzerland, Greece, the Netherlands, Belgium, Denmark, and Ireland.

Note 5: The Rest of Asia Pacific includes South Korea, Australia, and countries in Southeast Asia.

Note 6: The Rest of Latin America includes Argentina, Peru, Chile, and Colombia.

1.3.2 YEARS CONSIDERED



Note: For market estimates, the calendar year has been considered, whereas, for company profiles, the fiscal year has been considered.

1.4 CURRENCY

- The currency used in this report is the United States Dollar (USD), with market sizes indicated in terms of USD million/billion.
- For companies reporting their revenues in USD, the revenues were taken from their respective annual reports/SEC filings.
- For companies reporting their revenues in other currencies, the average annual currency conversion rate was used for that particular year to convert the value to USD.

1.5 LIMITATIONS

- The market study does not cover the market size in terms of volume, as it is a service-oriented market.
- A limited number of industry experts participated in primary research from Africa, the Middle East, and Latin America due to specific language barriers. In such cases, the geographic market size was derived based on weightages assigned to these markets based on the qualitative insights from global industry experts, the analysis of demographic parameters, and the adoption trends for microbiome sequencing observed in these markets.
- Financial information for privately held and unlisted companies was not available in the public domain. Hence, their financial details are not provided in the company profiles section.
- Company developments not reported in the public domain are not included in this report.
- Market shares are estimated on a year-on-year basis (2020 to 2022) using historical data of companies and secondary sources.
- Several strategic approaches and recommendations from industry experts were utilized to collate the research study to overcome the limitations and maintain the effectiveness of the study.

1.6 STAKEHOLDERS

- Research Institutes
- Pharmaceutical and Biotechnology Companies
- Microbiome Sequencing Service Providers
- Metagenomic Sequencing Service Providers
- NGS Service Providers
- Venture Capitalists and Investors
- Contract Research Organizations
- Government Associations
- Healthcare Associations/Institutes
- Business Research and Consulting Service Providers

1.7 SUMMARY OF CHANGES

Refinements in the segments of the global microbiome sequencing services market:

- Market sizes are updated for the base year 2022 and forecasted from 2023 to 2028.
- The new edition of the report provides updated financial information till 2022. This will help in the easy analysis of the present status of profiled companies in terms of their financial strength, profitability, key revenue-generating region/country, and business segment focus in terms of the highest revenue-generating segment.
- Recent developments are helpful in understanding market trends and the growth strategies adopted by players in the market. The number of service launches has increased in the last three years (October 2020 to April 2023).
- Tracking the service portfolios of prominent market players helps analyze the major services in the microbiome sequencing services market. The new edition of the report provides an updated service portfolio of the companies profiled in the report.
- Key market strategies, market share analysis, and competitive leadership mapping have been updated in the competitive landscape section of the report.
- The updated version of the report includes the revenue share analysis of key market players from 2020 to 2022.
- The current edition of the report includes an assessment of the economic and financial recession and geographical conditions such as supply chain disruptions, the Russia-Ukraine war, along with the effects of inflation on the services market. The impact of the recession on year-on-year growth during the forecast period (2023 to 2028) has also been considered.